Apartment Mid-Rise Prototype Scorecard Spreadsheet Review

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Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Many envelope values seem to reference the wrong source (APTM26, APTM28, APTM29, APTM30, APTM31, APTM32, APTM33, APTM34, APTM35, APTM36, and APTM40)
- The ventilation for retail is 0.2 cfm/ft2 not 0.15 (APTM48-Ventilation).
- Some hot water usage values incorrect (APTM58-Water Heater, APTM59-Water Heater).
- Interior lighting values for several space types (APTM66-Interior Lights, APTM67-Interior Lights, APTM72-Interior Lights).
- A few heating setpoint schedule values for residential units were copied wrong (APTM87-Schedules) and also for a few retail equipment hours (APTM88-Schedules)

The following are the highest priority comments:

- The multipliers on Zones tab for the dwelling units are used incorrectly resulting in the number of people being undercounted (APTM15-Zones, APTM16-Zones).
- The number of people for the dwelling units and many other spaces is too high (APTM17-Zones, APTM18-Zones, APTM38-Envelope).
- The fraction of single pane to double pane windows is used for adjusting some older vintage U-Factors and SHGC but is based on hotels instead of apartment buildings.
- Natural ventilation should be described since common in dwellings (APM52-Ventilation).
- A diversity of thermostat setpoints would capture the variation that exists for different families (APTM56-Thermostat)

- The total peak hot water usage of 116 gal/hr and storage volume of 47 gallons seems too small for a five-story mixed residential and retail building (APTM61-Water Heater)
- Cold water supply temperature of 55F should vary monthly and by climate zone (APTM65-Water Heater).
- Additional lighting power should be included for retail and other spaces (APTM68-Interior Lights).
- With retail and residential access on the first floor, three entrances is too low (APTM77-Ext Lights).

The following comments represent other important issues found:

- Shading geometry should be included since described in Ext Lighting tab (APTM9-Prototype).
- The office and mechanical rooms schedules should have their own schedules (APTM14-Zones).
- The parking garage area is larger than first floor or residential floor areas (APTM20-Zones)
- No space for storage lockers is included in the prototype (APTM23-Zones).
- No adjustment for thermal bridging is made (APTM24-Envelope).
- No internal window shades were assumed but is very common (APTM42-Envelope).
- For new construction, the residential tenant and common spaces should have different system descriptions (APTM45-HVAC System).
- The ventilation rate for residential units may be wrong due to incorrect multipliers (APTM50-Ventilation)
- It is very optimistic that all space for all older vintages would have updated their lighting system to be as efficient as Title 24-2016 (APTM71-Interior Lights).
- The use of stochastic schedules for the dwelling units would probably makes sense to capture the variation in occupancy (APTM89-Schedules, APTM90-Schedules).

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- What are the sources for window dimensions, shading geometry, floor heights, and sill height (APTM8-Prototype)?
- How are the number of studio, 1-, 2-, and 3-bedroom units determined (APTM13-Zones)?
- How is the fraction of operable windows determined? It is used to weight U-Factor and SHGC (APTM37-Envelope).

- How is the ACH50 calculated for the 2008-2018 vintage (APTM41-Envelope)?
- What is the heating efficiency for non-residential new construction spaces (APTM46-HVAC System)?
- How was the residential lighting power density determined (APTM70-Interior Lights)?
- What is the source for plug loads (APTM80-Equipment, APTM81-Equipment)?

Background

The review of the "scorecard," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Interior Lighting
- Ext Lighting
- Equipment
- Schedules

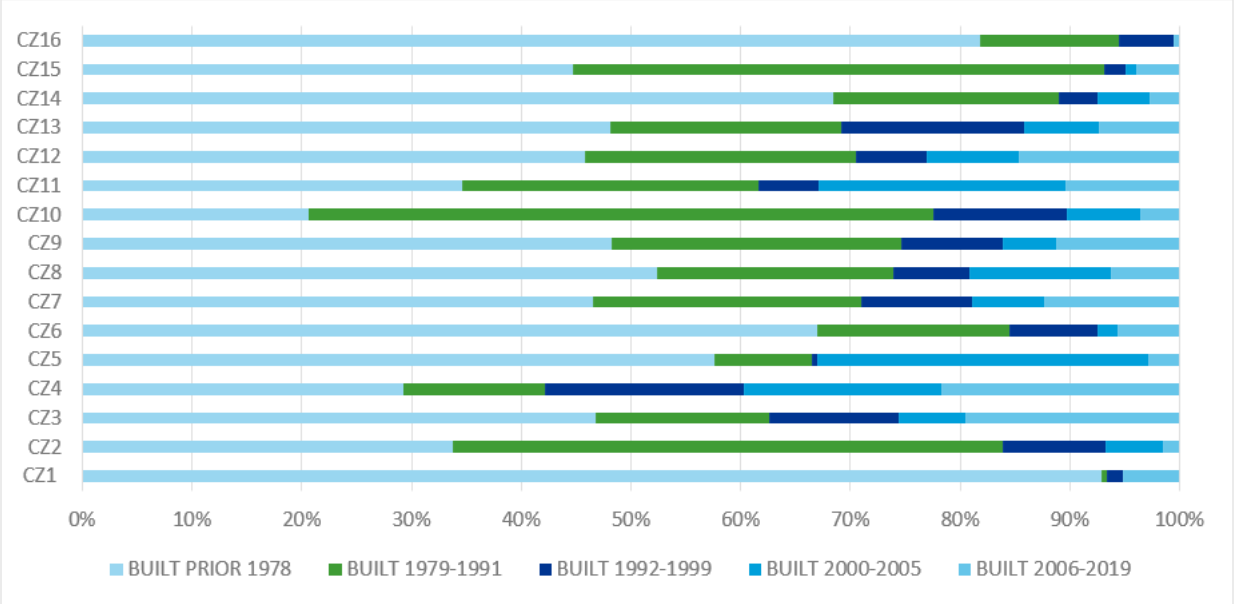
The comments on these tabs are labeled using the following format APTM-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

APTM1-Notes (B20:C30)

The Dodge data for apartment buildings of 4 or more stories may be supplemented with data from RASS shown in “DEER Residential EnergyPlus Prototypes: Development & Calibration” by DNV from 20222.

Figure 2-10 Cumulative distribution of vintages for all climate zones of multi-family homes



APTM2-Notes (n/a)

There are a number of major categories in multifamily buildings, but probably the categorization that would impact the energy use of the building is between Market Rate and Low-Income buildings. The physical size of the units, the building amenities, and the occupancy characteristics will probably be impacted by this differentiation. It may not be important but left unsaid in the supporting analysis; it implies that it was considered rather than it was not considered and the differences were not important.

Table A-2: Multifamily Project Number of Units by Income Category and Construction Start Period (2010-2012)

Units	Income Category	% of Projects per Time Period			
		2010	2011	2012	2010-2012
Unknown	Market-Rate *	23%	15%	14%	17%
	Low-Income	3%	0%	<1%	1%
3 to 4	Market-Rate	2%	4%	3%	3%
	Low-Income	0%	0%	<1%	<1%
5 to 19	Market-Rate	7%	10%	6%	7%
	Low-Income	3%	4%	1%	2%
20 to 39	Market-Rate	2%	8%	5%	5%
	Low-Income	9%	6%	10%	8%
40 to 59	Market-Rate	4%	5%	3%	4%
	Low-Income	9%	12%	18%	13%
60 to 79	Market-Rate	2%	<1%	2%	2%
	Low-Income	11%	10%	13%	11%
80 to 99	Market-Rate	3%	1%	1%	2%
	Low-Income	7%	4%	5%	5%
100 to 199	Market-Rate	4%	5%	5%	4%
	Low-Income	8%	8%	5%	7%
200 or more	Market-Rate	1%	8%	7%	5%
	Low-Income	3%	1%	1%	2%
Total Projects		**246**	**239**	**278**	**763**

Sources: (1) McGraw Hill Construction. Dodge project data for 2010-2012, *MHC Project List for State of CA Multi 12_12Bid Date.xls*, Received December 2012; (2) California Tax Credit Allocation Committee, 2009-2011 Annual Reports, http://www.treasurer.ca.gov/ctcac/annual_reports.asp, accessed June 2013.

* CTCAC awardee lists consistently included the number of units in a given project; however, nearly one-half of the 467 MHC projects (55%) did not have unit data. As a result, it is not surprising that low-income projects, of which the majority came from CTCAC data only, are notably more likely to have a known number of units than market-rate projects.

This is one of many tables extracted from "Final Phase I Report: Baseline Characterization Market Effects Study of Investor-Owned Utility Multifamily Residential New Construction Programs in California (2014)" by NMR Group (available on CALMAC).

APTM3-Notes (n/a)

Some of the data sources don't seem to have been used to set or inform any numbers or other assumptions. Please remove those sources that were not used; it is just confusing to

have unused references. For those data sources that are used, please provide a URL and confirm that it is working. Also, please take a look at sources described on other tabs and make sure they are all listed with working URLs.

APTM4-Notes (n/a)

To provide a thorough review of the prototype, I reviewed several great resources shown below prior to beginning my review. Originally, 79 resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the school prototype are the following:

- End-Use Load Profiles for the U.S. Building Stock - Methodology and Results of Model Calibration, Validation, and Uncertainty Quantification docs.nlr.gov/docs/fy22osti/80889.pdf
- 2014 Building America House Simulation Protocols - E. Wilson, C. Engebrecht Metzger, S. Horowitz, and R. Hendron, National Renewable Energy Laboratory. https://docs.nlr.gov/docs/fy14osti/60988.pdf
- DEER Residential EnergyPlus Prototypes: Development & Calibration https://cedars.cpuc.ca.gov/deer-resources/tools/energy-plus/file/2975/download/

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- Assessing the impact of occupant behavior on residential building performance: A case study of window operation, Farjadnia et al https://pdf.sciencedirectassets.com/271434/1-s2.0-S0360132326X20066/1-s2.0-S0360132326003586/main.pdf
- 2017 Characteristics of New Housing https://www.census.gov/construction/chars/pdf/c25ann2017.pdf
- Energy Code Field Studies: Low-Rise Multifamily Air Leakage Testing (2020) Bohac et al. https://www.energycodes.gov/sites/default/files/2021-07/LRMF_AirLeakageTesting_FinalReport_2020-07-06.pdf
- Multifamily Building Study Pilot (2022) EIA https://www.eia.gov/consumption/residential/pdf/Multifamily_Building_Study_Pilot.pdf
- Final Phase I Report: Baseline Characterization - Market Effects Study of Investor-Owned Utility Multifamily Residential New Construction Programs in California

(2014) NMR Group CALMAC CPU0089.01
https://www.calmac.org/publications/WO54_MFNC_-_Phase_1_Report_Final_070814_.pdf

APTM5-Prototype (A7:C17)

The referenced "TRC Multifamily Prototypes study (2019)" that is used to justify the shape, floor area, aspect ratio, number of floors, roof type, window-to-wall ratio, window dimensions, shading geometry, floor and window height does not have a link or a proper reference. The link should be:

https://title24stakeholders.com/wp-content/uploads/2019/06/SCE-MFModeling_MultifamilyPrototypesReport_2019-06-07_clean.pdf

This should appear on the Notes tab. Since this is a highly referenced report, it should probably also appear in the scorecard "sources" folder on SharePoint and in the "documentation" folder on the CalBEM-Prototypes GitHub.

APTM6-Prototype (A7:C17)

The referenced "TRC Multifamily Prototypes study (2019)" that is used to justify many major building parameters may be biased toward representing newer buildings based on its data sources that are for buildings from 2014 to newer.

	CalCERTS HERS Registry	California Multifamily New Homes Program	CoStar	Project lists shared by consultants
Region	California statewide	PG&E service territory	California statewide	California statewide
Years built	2015-2017	2015-2017	2014-2018	2014-2018
Low-rise/high-rise	Low-rise	Low-rise and high-rise	Low-rise and high-rise	Low-rise and high-rise
No. properties	unknown	60	1,395	95
No. buildings	744	unknown	3,915	273
No. dwelling Units	10,768	3,832	164,504	10,064

Figure 2: Summary of data sources

APTM7-Prototype (B12)

The referenced “TRC Multifamily Prototypes study (2019)” that is used to justify many major building parameters shows the mid-rise mixed-use prototype window-to-wall ratio is 25%, not the 22% shown in the scorecard.

	Low-Rise Garden Style	Loaded Corridor	Mid-Rise Mixed Use	High-Rise Mixed Use
Total floor area	7,320	39,372	113,100	125,400
Window area needed to meet four percent window opening to floor ratio	586	3,150	9,048	10,032
Total exterior wall area	5,472	14,094	39,836	46,592
Window to Wall Ratio	11%	22%	23%	22%
WWR Range from consultant data	4 to 26%	4 to 20%	5 to 40%	24 to 66%
Average WWR from consultant data	15%	9%	20%	41%
Prototype WWR	**15%**	**25%**	**25%**	**40%**

Figure 10: Window-to-wall ratio (WWR) by prototype (Source: consultant project lists)

Perhaps the 25% is intended for the residential floors and a different WWR is for the first floor. If that is the case, please explain that in the scorecard.

APTM8-Prototype (A12:B17)

The following parameters are not justified based on the TRC study, and if they are from the CEC Prototype, they should be further justified:

- Window dimensions: 5x3
- Shading Geometry: none
- Floor-to-floor height (feet): 10 for residential/non-residential, 15 for parking
- Floor-to-ceiling height (feet): 9 for residential, 10 for non-residential, 14 for parking
- Glazing sill height (feet): 3.1 ft first floor, 2.95 ft upper floors

APTM9-Prototype (A14:B14)

The shading geometry is shown as a dash, which implies no shading, but on the “Ext Lighting” tab, a sales canopy is shown, which would provide shading. These should be consistent.

APTM10-Prototype (n/a)

The number of apartments is a key factor for a multifamily building and should be listed as a major assumption. Providing the number of studio, 1-, 2-, and 3-bedroom units on this tab, while not as key, would provide a good understanding of the building.

APTM11-Prototype (n/a)

It would be good to compare multiple prototypes that exist for this building category to contrast the major assumptions between them and this prototype. This has been done with other scorecards and is useful.

APTM12-Zones (A15:B16)

It is unlikely that for parking the floor-to-floor height (shown as 15 feet) would be different than the floor-to-ceiling height (shown as 14 feet). The thickness of the floor is not considered in building energy models, so this implies a 1 ft high plenum area above parking, which is very unusual.

APTM13-Zones (T1:W17)

Each residential floor in the building, according to the plans, has:

- 2 studio units (9%)
- 10 one-bedroom units (45%)
- 8 two-bedroom units (36%)
- 2 three-bedroom units (9%)

It is not clear where the fraction of each apartment type data came from. This is close to the statistics in “2017 CHARACTERISTICS OF NEW HOUSING” (U.S. Department of Commerce) shown below:

Number of Multifamily Units Completed by Number of Bedrooms per Unit

(Components may not add to totals because of rounding. Percents computed from unrounded figures.)

Year	Number of units (in thousands)					Percent distribution				
	Total	Efficiencies	1 bedroom	2 bedrooms	3 or more bedrooms	Total	Efficiencies	1 bedroom	2 bedrooms	3 or more bedrooms
	West									
1978	160	8	52	84	16	100	5	32	52	10
1979	169	6	51	92	20	100	3	30	55	12
1980	153	10	42	83	19	100	6	27	54	12
1981	111	4	31	63	14	100	3	28	57	12
1982	82	2	25	47	8	100	2	30	58	10
1983	104	3	34	58	9	100	3	33	56	9
1984	164	5	56	90	12	100	3	34	55	7
1985	207	6	71	117	14	100	3	34	56	7
1986	216	6	70	124	16	100	3	32	57	7
1987	190	6	57	110	17	100	3	30	58	9
1988	156	4	51	88	13	100	3	33	57	8
1989	131	4	42	72	14	100	3	32	55	11
1990	121	3	37	64	18	100	2	30	53	15
1991	87	1	21	52	12	100	2	25	60	13
1992	58	1	15	35	7	100	2	25	60	13
1993	43	1	9	24	8	100	3	22	56	19
1994	50	2	12	27	10	100	4	24	54	19
1995	63	1	14	36	12	100	2	23	57	19
1996	77	1	21	42	14	100	1	27	54	18
1997	77	3	21	40	14	100	4	27	52	18
1998	78	2	23	39	14	100	3	30	50	17
1999	85	1	25	41	17	100	2	30	48	20
2000	77	4	25	35	14	100	5	33	45	18
2001	80	2	26	37	15	100	3	32	46	19
2002	88	3	30	39	17	100	3	34	44	19
2003	73	4	22	32	16	100	5	29	43	22
2004	74	3	23	34	14	100	4	31	46	19
2005	69	1	18	35	14	100	2	26	51	21
2006	74	2	21	36	15	100	3	28	49	21
2007	75	2	20	38	16	100	3	26	50	21
2008	75	5	22	33	15	100	7	29	45	19
2009	69	3	26	32	9	100	4	37	46	13
2010	45	5	16	18	5	100	12	36	41	12
2011	23	1	8	9	5	100	6	33	40	21
2012	39	3	14	15	7	100	8	35	39	18
2013	50	4	18	20	8	100	8	36	40	16
2014	68	6	26	26	9	100	9	39	38	14
2015	85	5	35	35	10	100	6	41	41	12
2016	82	6	31	35	9	100	8	38	43	11
2017	90	9	34	36	11	100	10	38	40	13
RSE/SE	7	13	8	8	17	(NA)	1	2	2	2

Footnotes:
RSE/SE - Relative Standard Error (percent), Standard Error (percentage points)
NA - Not available
A - Represents an RSE or SE that is greater or equal to 100 percent or could not be computed
Z - Less than 500 units or less than 0.5 percent
S - Withheld because estimate did not meet publication standards on the basis of response rate or a consistency review

These values for more recent years are available here and do not show a significant change in the statistics:

https://www.census.gov/construction/chars/current.html

But earlier years show fewer studio (efficiency) apartments and a higher fraction of two-bedroom units.

APTM14-Zones (D6, D8)

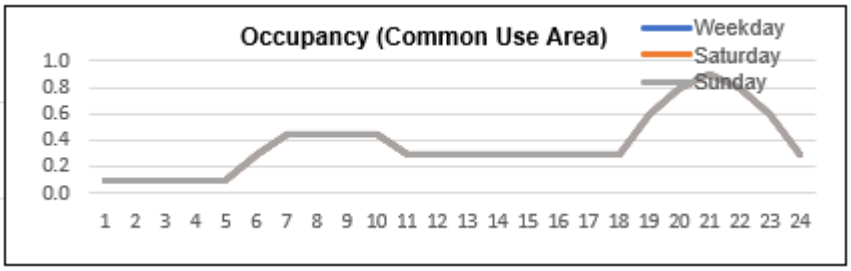
The scorecard describes Schedule Categories of:

- ResidentialCommon
- Retail
- ResidentialLiving

Unfortunately, while most spaces in the building can be categorized into these, two spaces do not fit well:

- Office Area
- Electrical, Mechanical, Telephone Rooms

The office is the leasing office and would likely have normal or perhaps slightly longer than normal office hours, not the hours one would expect in a corridor:



APTM15-Zones (E2, E13:E38)

The number and accounting for the living spaces is confusing on the Zones tab, which in cell E40 shows "37 dwelling units" even though the TRC Multifamily Prototypes study (2019) and the diagram on the tab show 22 dwelling units per floor times four floors, which equals 88 dwelling units. Part of the reason for the confusion is that the area of the following zones represents an implied multiplier of 2 to 5, depending on the unit:

- F2-4 1-BED CORE EAST
- F2-4 1-BED CORE N
- F2-4 1-BED CORE WEST

- F2-4 1-BED NORTH
- F2-4 2-BED CORE NE-NW
- F2-4 2-BED EAST
- F2-4 2-BED WEST

When this adjustment is made, each floor shows 22 units as expected. I would recommend using the areas from each individual dwelling unit, then a multiplier for the number of those dwelling units on each floor, and then a floor multiplier is applied to that. This is further confused by the heading of "No. of Dwelling Unit" in column E and the "Multiplier" in column L being the same number (and actually linked by a simple formula).

APTM16-Zones (Column M)

The areas and multiplier shown are inconsistent with the people column calculations. The people column has the formula:

=N13*J13/1000

This only represents the area of the thermal zone multiplied by the persons per 1000 ft2 but does not include the multiplier for the number of stories. This is obvious when looking at a corresponding apartment for floors 2-4 and floor 5. The zone "F2-4 3-BED NW" has 1410 sqft, and the "F5 3-BED NW" has the same 1410 sqft, and they have the same number of people: 7. The row with "F2-4 3-BED NW" should be multiplied by 3 to have 21 total people, but it isn't. When the multiplier is applied correctly, the number of people jumps to 1050 (see column Z in my copy of the spreadsheet).

APTM17-Zones (Column M)

The number of people shown in the apartments is 199, which seems a bit high for the mix of apartments with different numbers of bedrooms (shown per floor):

- 2 studio units
- 10 one-bedroom units
- 8 two-bedroom units
- 2 three-bedroom units

Presumably, most studios are occupied by an individual, one-bedroom units are typically occupied by one or two individuals, two-bedroom units by 2 to 3 individuals, and three-bedroom units by 3 to 4 individuals.

- 2 studio units x 1 people
- 10 one-bedroom units x 1.5 people
- 8 two-bedroom units x 2.5 people
- 2 three-bedroom units x 3.5 people

This totals 44 people/floor or 176 people overall.

Another source, "2014 Building America House Simulation Protocols," uses the number of bedrooms as a proxy and has the formula:

Number of occupants = 0.92 x Bedrooms + 0.63

Calculating this for each dwelling unit:

- 2 studio units x 0.63 people (not sure that less than one makes sense)
- 10 one-bedroom units x 1.55 people
- 8 two-bedroom units x 2.47 people
- 2 three-bedroom units x 3.39 people

Would result in 43.3 people/floor or 173.2 people overall.

While the original total for the number of people in residential living places is 199, after the formula is fixed, it grows to 380 (see APTM16-Zones**).** In order to get to those numbers, the quantity of people in each apartment would need to more than double:

- 2 studio units x 2.2 people
- 10 one-bedroom units x 3.2 people
- 8 two-bedroom units x 5.4 people
- 2 three-bedroom units x 7.6 people

These quantities are simply unrealistic as the average number of people in these apartments.

APTM18-Zones (Column M)

Based on the previous comments (APTM15-Zones , APTM16-Zones), after fixing the formula for the number of people, the number of people per space type is shown below:

- Convention, Conference, Multipurpose and Meeting Area 24
- Corridor Area 119
- Exercise/Fitness Center and Gymnasium Areas 18

- Office Area (>250 square feet) 3
- Lobby, Main Entry 70
- Electrical, Mechanical, Telephone Rooms 1
- Retail Sales Area (Retail Merchandise Sales) 292
- High-Rise Residential Living Spaces 380
- Laundry Area 4
- Parking Garage Area (Parking Zone and Ramps) 139

A previous comment focused on the apartment occupancy (APTM17-**Zones**), but some of these other space types seem equally too high (shown highlighted).

Even adjusting for schedules, which are at 90% at their peak at 9 pm, the corridors have 107 people, the lobby has 63, and the garage has 125. The retail peaks on Saturday afternoon at 80%, which is 233 people in 17,615 ft2.

Can anyone say they have seen more than a dozen people in an apartment building parking lot at any given time? Not 125. And what about the lobby of an apartment building? 63 people is probably double the number of seats, even in a large apartment building lobby. Since the residential-living schedule peaks at 70% at the same time as the residential-common schedule is at 90% at 9 pm, this means that 606 people are in the building at that time, which is more than double the number of people in apartments at that time 266. That is a lot of guests at 9 pm every night in an apartment building.

This quantity of people may be driven by using the maximum occupancy for ventilation calculations instead of a realistic number of people for spaces.

APTM19-Zones (A13:A21, A24:A34)

The term “High-Rise Residential Living Spaces” seems like a poor description of a space type in a mid-rise apartment building. Perhaps just call it “Residential Living Spaces.”

APTM20-Zones (J39)

The area of the Parking Garage Area (Parking Zone and Ramps) space type called the PARKING GARAGE space/thermal zone is 27,903 ft2, but the area of the first floor and residential floors are each 22,622 ft2. That is over 5000 ft2 larger, which seems unlikely since it is rare for a building to be constructed with a larger basement than the first floor. Some area for ramps might be appropriate, but 5000 ft2 seems too large for this.

APTM21-Zones (J39)

While the number of parking spaces is not explicitly listed, according to several websites, including:

https://adainspectionorangecounty.com/california-apartment-parking-laws/

Many California cities have specific requirements for apartment building parking

- Studios and 1-bedroom: 1 space per unit
- 2-bedroom: 1.5 spaces per unit
- 3-bedroom or larger: 2 spaces per unit

For each of the four residential floors, this prototype has

- 2 studio units
- 10 one-bedroom units
- 8 two-bedroom units
- 2 three-bedroom units

For this building, with 88 units, that would require 112 parking spaces. For garages, 320 ft2 to 350 ft2 for each parking space is typical, which would result in 35,840 to 39,200 ft2 needed. This is much larger than the 27,903 ft2 shown as parking garage area for the prototype. This means that either a parking lot is supplementing the garage area or that the garage should be larger, perhaps two underground floors. This should be coordinated with the exterior lighting requirement for parking lots.

APTM22-Zones (A9:A12)

The inclusion of retail spaces in the building does not seem consistent with EIA's "Multifamily Building Study Pilot: Methods, Findings, and Lessons Learned (2022)" study:

Table 8. Prevalence of commercial space by geography

Characteristic	Geography									
	Combined		New York		California		Harris County, Texas		Denver Metropolitan Statistical Area	
	N	%	N	%	N	%	N	%	N	%
Commercial space										
Yes	34	12.7%	22	25.3%	6	6.5%	2	4.8%	4	8.7%
No	234	87.3%	65	74.7%	87	93.5%	40	95.2%	42	91.3%
Percentage commercial										
≤ 5%	11	36.7%	10	52.6%	1	20.0%	0	0.0%	0	0.0%
6%–10%	7	23.3%	2	10.5%	3	60.0%	1	50.0%	1	25.0%
> 10%	12	40.0%	7	36.8%	1	20.0%	1	50.0%	3	75.0%

Data source: U.S. Energy Information Administration, Multifamily Building Study

Which shows commercial space as being unusual in California.

APTM23-Zones (A9:A12)

Storage lockers, sometimes allocated by residence or associated with additional rent, are common in apartment buildings. Usually, storage lockers are located in a basement or in the core of a floor that has storage rooms without windows.

APTM24-Envelope (n/a)

No thermal bridging adjustments are being described in the scorecard. For other studies, especially looking at older buildings, some assumptions of how thermal bridges should be accounted for in the model are provided. It is likely that thermal bridging mitigation is only present for newer vintages. In buildings such as multifamily buildings where the internal loads are generally low, the impact of thermal bridging can be a significant fraction of the total heating and cooling loads.

APTM25-Envelope (Column A)

The indication “F1” in “External Wall (F1)” and “Windows_Nonres (F1)” as an acronym for the first floor is not very clear. Please just say “Floor 1” instead. For “External Wall” and “Windows_Dwelling_Units,” please indicate what floors they apply to (2 through 5). Perhaps just adding a column “Applies to Floors” with the appropriate values would make sense.

APTM26-Envelope (I3:Y3)

The “pre-1978” vintage for the first-floor wall, which is described in the note as ASHRAE “90.1, 2022, Table A3.3.3.1, 3.5, R-0” and the U-factor of 0.352, is consistent with this table, but unfortunately, the referenced table describes steel-frame walls and not concrete walls.

Table A3.3.3.1 Assembly U-Factors for Steel-Frame Walls

Framing Type and Spacing Width (Actual Depth)	Cavity Insulation R-Value: Rated (Effective Installed [see Table A9.2-2])	Overall U-Factor for Entire Base Wall Assembly	Overall U-Factor for Assembly of Base Wall Plus Rated R-Value of										
			R-1.00	R-2.00	R-3.00	R-4.00	R-5.00	R-6.00	R-7.00	R-8.00	R-9.00	R-10.00	R
Steel Framing at 16 in. on Center													
3.5 in. depth	None (0.0)	0.352	0.260	0.207	0.171	0.146	0.128	0.113	0.102	0.092	0.084	0.078	

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The basis for assuming a concrete construction comes from the “Construction” column, which states “8-in cast-in-place concrete wall + insulation + air/water barrier + exterior cladding + gypsum board”

The appropriate table from 90.1-2022 is Table A3.1-1, which shows a U-factor of 0.414 for solid concrete walls with R-0 but interior gypsum board.

Table A3.1-1 Assembly U-Factors for Above-Grade Concrete Walls and Masonry Walls

Framing Type and Depth	Rated R-Value of Insulation Alone	Assembly U-Factors for 8 in. Normal Weight 145 lb/ft^3 Solid Concrete Walls	Assembly U-Factors for 8 in. Medium Weight 115 lb/ft^3 Concrete Block Walls: Solid Grouted	Assembly U-Factors for 8 in. Medium Weight 115 lb/ft^3 Concrete Block Walls: Partially Grouted (Cores Uninsulated Except Where Specified)
	R-0	U-0.740	U-0.580	U-0.480
No Framing	Ungrouted Cores Filled with Loose-Fill Insulation	NA	NA	U-0.350
Continuous Metal Framing at 24 in. on Center Horizontally				
1.0 in.	R-0	U-0.414	U-0.359	U-0.318

The basis for assuming it is concrete on the first floor only comes from the term “External Wall (F1)” under the “Element” column, where F has been used in other portions of the spreadsheet to indicate the Floor number. In addition, the “TRC Multifamily Prototypes study (2019)” referenced on the Prototype tab shows the following:

	Low-Rise Garden Style	Low-Rise Loaded Corridor	Mid-Rise Mixed-Use	High-Rise Mixed-Use
Stories	2	3	5 (1 commercial, 4 residential)	10 (1 commercial, 9 residential)
No. dwelling units	**8** 2 1-bedroom 2 2-bedroom	**36** 6 studio 12 1-bedroom 12 2-bedroom 6 3-bedroom	**88** 8 studios 40 1-bedroom 32 2-bedroom 8 3-bedroom	**117** 18 studios 54 1-bedroom 45 2-bedroom
Conditioned floor area	7,320	39,372	113,700	125,400
Foundation	Slab on grade	Slab on grade	Concrete podium with underground parking	Concrete podium with underground parking
Wall assembly	Wood frame	Wood frame	Wood frame over a first floor concrete podium	Steel frame
Roof assembly	Low slope attic roof	Flat roof	Flat roof	Flat roof
Window-to-wall ratio	15 percent	25 percent	25 percent	40 percent
Space heating and cooling	Individual ducted split heat pump	Individual ducted split heat pump	Individual ducted split heat pump	Four-pipe fan coil
Ventilation	Exhaust only	Exhaust only	Exhaust only	Central supply ventilation ducted to corridors and units
Domestic hot water	Individual gas instantaneous	Gas storage serving multiple units	Gas storage serving whole-building	Gas storage serving whole-building

Figure 1: Table summary of proposed prototype characteristics

APTM27-Envelope (I9)

The cell shows “Title 24 1984 Table 2-53 BB, wall density less than 25,” but it should say “Title 24 1986”. That is where the values are from and is the version referenced under the “Selected T24” column.

APTM28-Envelope (I5:Y5)

For the 1998-2007 vintage, the Title 24-2001 version was selected for concrete walls used on the first floor, but the values from Table 1-H being used are for metal-framed walls instead of concrete “mass” walls. The U-values in the scorecard are all 0.182 or 0.189.

TABLE 1-H—PRESCRIPTIVE ENVELOPE CRITERIA
FOR NONRESIDENTIAL BUILDINGS
(Except high-rise residential buildings and guest rooms of hotel/motel buildings)

	CLIMATE ZONES				
	1, 16	**~~23~~-5**	**6-~~10~~9**	**2, 10~~11~~-13**	**14, 15**
Roof/Ceiling					
R-value or	19	19	11	19	19
~~U-value~~U-factor	0.057	0.057	0.078	0.057	0.057
Wall					
R-value or	13	11	11	13	13
~~U-value~~U-factor					
Wood frame	0.084	0.092	0.092	0.084	0.084
Metal frame	0.182	0.189	0.189	0.182	0.182
Mass/7.0≤ HC<15.0	0.340	0.430	0.430	0.430	0.430
Mass/15.0≤HC	0.360	0.650	0.690	0.650	0.400
Other	0.084	0.092	0.092	0.084	0.084
Floor/Soffit					

APTM29-Envelope (I10, J10, Y10)

The values for the 1998-2007 vintage for the four residential stories that are wood-framed are cited and taken from Table 1-H, but Table 1-I is more appropriate for residential buildings:

TABLE 1-H—PRESCRIPTIVE ENVELOPE CRITERIA
FOR NONRESIDENTIAL BUILDINGS
(Except high-rise residential buildings and guest rooms of hotel/motel buildings)

TABLE 1-I—PRESCRIPTIVE ENVELOPE CRITERIA FOR HIGH-RISE RESIDENTIAL BUILDINGS AND GUEST ROOMS OF HOTEL/MOTEL BUILDINGS

HIGH-RISE RESIDENTIAL BUILDING is a building, other than a hotel/motel, of Occupancy Group R, Division 1 with four or more habitable stories.

Due to the similar criteria in the two tables for CZ2-CZ15, the only two values that are incorrect are from CZ1 and CZ16, as well as the citation to the wrong table.

APTM30-Envelope (I25:Y25)

The 1998-2007 vintage that uses T24-2001 has rows for SHGC that are dependent on the amount of glazing expressed as window-to-wall ratio. The building, according to the Prototype tab, is 22% glazed, while the cited study, "TRC Multifamily Prototypes study (2019)," shows the residential floors as 25% glazed. The values for SHGC are taken from the 31% to 40% row of the table instead of the 21% to 30% row, so they are incorrect except for CZ14 and CZ15, which happen to have the same values for both window-to-wall ratio categories.

TABLE 1-I—PRESCRIPTIVE ENVELOPE CRITERIA FOR HIGH-RISE RESIDENTIAL BUILDINGS AND GUEST ROOMS OF HOTEL/MOTEL BUILDINGS

	CLIMATE ZONES									
	1, 16		~~2~~3-5		6-~~10~~9		~~11~~2,10-13		14, 15	
Windows										
~~U-value~~U-factor	~~0.72~~0.49		~~1.23~~0.49		~~1.23~~0.49		~~0.72~~0.49		~~0.72~~0.49	
Relative solar heat gain										
~~North~~	~~0.77~~		~~0.82~~		~~0.82~~		~~0.77~~		~~0.77~~	
~~Nonnorth~~	~~0.77~~		~~0.82~~		~~0.62~~		~~0.50~~		~~0.50~~	
	Non-North	North	Non-North	North	Non-North	North	Non-North	North	Non-North	North
0-10% WWR	0.46	0.68	0.41	0.61	0.47	0.61	0.36	0.49	0.36	0.47
11-20% WWR	0.46	0.68	0.40	0.61	0.40	0.61	0.36	0.49	0.31	0.43
21-30% WWR	0.36	0.47	0.31	0.61	0.36	0.61	0.31	0.40	0.26	0.43
31-40% WWR	0.30	0.47	0.26	0.55	0.31	0.61	0.26	0.40	0.26	0.31

APTM31-Envelope (L30:CZ30)

For the 1998-2007 vintages from Title 24-2001, for the non-residential floor of the buildings, Table 1-H should be the source of the values, but for CZ3 to CZ9, the values seem to be coming from Table 1-I for residential floors instead of 1H.

APTM32-Envelope (I35:Y35)

Like in comment APTM30-Envelope, the non-residential SHGC for the 1998-2007 vintage is using the 31-40% WWR instead of the 21-30%. Most values are incorrect due to this, except for CZ1, CZ3-CZ5, CZ16.

APTM33-Envelope (J6:Y6)

As in comment APTM28-Envelope, the values for the concrete non-residential floor are using metal-framed instead of the mass-heavy row of Table 140.3-B of Title 24-2013 representing the 2008-2018 vintage.

TABLE 140.3-B – PRESCRIPTIVE ENVELOPE CRITERIA FOR NONRESIDENTIAL BUILDINGS (INCLUDING RELOCATABLE PUBLIC SCHOOL BUILDINGS WHERE MANUFACTURER CERTIFIES USE ONLY IN SPECIFIC CLIMATE ZONE; NOT INCLUDING HIGH-RISE RESIDENTIAL BUILDINGS AND GUEST ROOMS OF HOTEL/MOTEL BUILDINGS)

				Climate Zone															
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	Maximum U-factor	Roofs/ Ceilings	Metal Building	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065
			Wood Framed and Other	0.049	0.039	0.039	0.039	0.049	0.075	0.067	0.067	0.039	0.039	0.039	0.039	0.039	0.039	0.039	0.039
		Walls	Metal Building	0.113	0.061	0.113	0.061	0.061	0.113	0.113	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.057	0.061
			Metal-framed	0.098	0.062	0.082	0.062	0.062	0.098	0.098	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062
			Mass Light[1]	0.196	0.170	0.278	0.227	0.440	0.440	0.440	0.440	0.440	0.170	0.170	0.170	0.170	0.170	0.170	0.170
			Mass Heavy[1]	0.253	0.650	0.650	0.650	0.650	0.690	0.690	0.690	0.690	0.650	0.184	0.253	0.211	0.184	0.184	0.160
			Wood-framed and Other	0.102	0.059	0.110	0.059	0.102	0.110	0.110	0.102	0.059	0.059	0.059	0.059	0.059	0.059	0.042	0.059

APTM34-Envelope (J7:Y7)

Like in comment APTM28-Envelope, the values for the concrete non-residential floor are using metal-framed instead of the mass-heavy row of Table 140.3-B of Title 24-2025 representing the 2019+ vintage.

APTM35-Envelope (J12:Y12)

The wood-frame wall values for multifamily for the 2019+ vintage from Title 24-2025 seem to be using the < 1-hr fire-rated values, but the description is for the >1 hr fire rated values. This means that 10 of the 16 climate zone values are incorrect.

TABLE 170.2-A ENVELOPE COMPONENT PACKAGE – Multifamily Standard Building Design (continued)

Building Component - Walls, Floors, Doors, and QII	CZ 1	CZ 2	CZ 3	CZ 4	CZ 5	CZ 6	CZ 7	CZ 8	CZ 9	CZ 10	CZ 11	CZ 12	CZ 13	CZ 14	CZ 15	CZ 16
Walls - Metal Building - Any Fire Rating	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.057	0.057	0.057	0.057	0.057	0.057
Walls - Framed (wood, metal) and other - >1hr fire rating	0.059	0.059	0.059	0.059	0.059	0.065	0.065	0.059	0.059	0.059	0.051	0.059	0.059	0.051	0.051	0.051
Walls - Framed (wood, metal) and other - ≤1hr fire rating[3]	0.051	0.051	0.051	0.051	0.051	0.065	0.065	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051

APTM36-Envelope (J6:Y6)

Like in comment APTM28-Envelope, the wood-framed values for the 2008-2018 vintage using Title 24-2013 should be using “Mass Heavy” criteria and not the “Metal-framed” criteria.

TABLE 140.3-B – PRESCRIPTIVE ENVELOPE CRITERIA FOR NONRESIDENTIAL BUILDINGS (INCLUDING RELOCATABLE PUBLIC SCHOOL BUILDINGS WHERE MANUFACTURER CERTIFIES USE ONLY IN SPECIFIC CLIMATE ZONE; NOT INCLUDING HIGH-RISE RESIDENTIAL BUILDINGS AND GUEST ROOMS OF HOTEL/MOTEL BUILDINGS)

				Climate Zone															
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	Maximum U-factor	Roofs/ Ceilings	Metal Building	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065
			Wood Framed and Other	0.049	0.039	0.039	0.039	0.049	0.075	0.067	0.067	0.039	0.039	0.039	0.039	0.039	0.039	0.039	0.039
		Walls	Metal Building	0.113	0.061	0.113	0.061	0.061	0.113	0.113	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.057	0.061
			Metal-framed	0.098	0.062	0.082	0.062	0.062	0.098	0.098	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062
			Mass Light[1]	0.196	0.170	0.278	0.227	0.440	0.440	0.440	0.440	0.440	0.170	0.170	0.170	0.170	0.170	0.170	0.170
			Mass Heavy[1]	0.253	0.650	0.650	0.650	0.650	0.690	0.690	0.690	0.690	0.650	0.184	0.253	0.211	0.184	0.184	0.160
			Wood-framed and Other	0.102	0.059	0.110	0.059	0.102	0.110	0.110	0.102	0.059	0.059	0.059	0.059	0.059	0.059	0.042	0.059

APTM37-Envelope (I21:Y21, I26:Y26)

The fenestration U-factor and SHGC for the 2008-2018 vintage using Title 24-2013 are based on the scorecard from the table below, assuming that 75% of the windows are fixed and the other 25% are operable. This is a key assumption that needs to be justified. The 25% seems very low to me since most midrise apartment buildings that I am familiar with are 100% operable windows in the dwelling units and only fixed windows in corridors or other common spaces.

CONTINUED: TABLE 140.3-C – PRESCRIPTIVE ENVELOPE CRITERIA FOR HIGH-RISE RESIDENTIAL BUILDINGS AND GUEST ROOMS OF HOTEL/MOTEL BUILDINGS

				All Climate Zones				
					Fixed Window	Operable Window	Curtainwall/ Storefront	Glazed Doors
Envelope	nestration	Vertical	Area-Weighted Performance Rating	Max U-factor	0.36	0.46	0.41	0.45
				Max RSHGC	0.25	0.22	0.26	0.23
			Area-Weighted Performance Rating	Min VT	0.42	0.32	0.46	0.17
			Maximum WWR%	40%				

APTM38-Envelope (J22:Y22)

The U-factors and SHGC for fenestration for the residential floors for the 2019+ vintage from Title 24-2025 should be based on the table below for “all other fenestration,” but the cells contain a formula that seems to be related to operable and non-operable values, which do not exist in Title 24-2025 for multifamily buildings.

TABLE 170.2-A ENVELOPE COMPONENT PACKAGE – Multifamily Standard Building Design (continued)

Building Component - Fenestration	CZ 1	CZ 2	CZ 3	CZ 4	CZ 5	CZ 6	CZ 7	CZ 8	CZ 9	CZ 10	CZ 11	CZ 12	CZ 13	CZ 14	CZ 15	CZ 16
Curtain Wall/ Storefront[7] - Maximum U-factor	0.38	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.38
Curtain Wall/ Storefront[7] - Maximum RSHGC	NR	0.26	NR	0.26	NR	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.25	0.26	NR
Curtain Wall/ Storefront[7] - Minimum VT, common use area	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46
NAFS 2017 Performance Class AW[5] - Maximum U-factor	0.38	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.38
NAFS 2017 Performance Class AW[5] - Maximum RSHGC	NR	0.24	NR	0.24	NR	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	NR
NAFS 2017 Performance Class AW[5] - Minimum VT, common use areas	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37
All Other Fenestration - Maximum U-factor	0.28	0.30	0.28	0.28	0.28	0.34	0.34	0.30	0.30	0.30	0.28	0.30	0.28	0.28	0.28	0.28
All Other Fenestration - Maximum RSHGC	NR	0.23	NR	0.23	NR	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	NR
Maximum Window to Floor Ratio	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%
Maximum Window to Wall Ratio	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%
Maximum Skylight Roof Ratio	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%

APTM39-Envelope (D18:D19, I18:Y19, D23:D24, I23:Y24, D28:D29, I28:Y29, D33:D34, I33:Y34)

The fenestration U-factors and SHGC for pre-1978 and 1978-1997 vintages for the residential and non-residential windows are based on the following comment in column D: “CBECS 2018 shows that ~58% of Large Hotels that were built before 2000 are single pane,” which is not statistically relevant to multifamily buildings. The formulas in columns I to Y are based on this ratio.

APTM40-Envelope (I3:Y37)

To summarize the above comments on the envelope criteria used in the scorecard, the following shows correct values in green and incorrect values in tan.

Element	Assembly Type	Construction	Thermal Properties	Vintage	Selected T24	Note	CZ1	CZ2	CZ3	CZ4	CZ5	CZ6	CZ7	CZ8	CZ9	CZ10	CZ11	CZ12	CZ13	CZ14	CZ15	CZ16
External Wall (F1)	ExtWall_MassLight_Nonres	8-in cast-in-place concrete wall + insulation + air/water barrier + exterior cladding + gypsum board	U-factor IP	Pre 1978	N/A	ASHRAE 90.1, 2022, Table A3.3.3.1, 3.5", R-0	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352	0.352
				1978-1997	1986	Copied from High-rise office alternative requirements (Table 2-53W): Package A with heat capacity around 10 (Btu/F-ft²)	0.213	0.227	0.227	0.217	0.217	0.256	0.256	0.256	0.256	0.256	0.238	0.238	0.238	0.208	0.208	0.213
				1998-2007	2001	Table 1-H	0.182	0.182	0.189	0.189	0.189	0.189	0.189	0.189	0.189	0.182	0.182	0.182	0.182	0.182	0.182	0.182
				2008-2018	2013	Table 140.3-B	0.098	0.062	0.082	0.062	0.062	0.098	0.098	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062	0.062
				2019+	2025	Table 140.3-B (Nonres)	0.06	0.055	0.071	0.055	0.055	0.06	0.06	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
External Wall	ExtWall_WoodFramed_Res	Wood-frame walls + insulation + Gypsum board	U-factor IP	Pre 1978	N/A	ASHRAE 90.1 Table A3.4.3.1, 3.5", R-0	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292	0.292
				1978-1997	1986	Title 24 1984 Table 2-53 BB, wall density less than 25	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095
				1998-2007	2001	TABLE 1-H	0.084	0.084	0.092	0.092	0.092	0.092	0.092	0.092	0.092	0.084	0.084	0.084	0.084	0.084	0.084	0.084
				2008-2018	2013	TABLE 140.3-B	0.102	0.059	0.119	0.059	0.102	0.119	0.110	0.102	0.059	0.059	0.059	0.059	0.059	0.042	0.059	0.059
				2019+	2025	TABLE 170.2-A ENVELOPE COMPONENT PACKAGE – Multifamily Standard Building Design Assuming - > 1hr fire rating	0.051	0.051	0.051	0.051	0.051	0.065	0.065	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051
Roof	FlatRoof_WoodFramedAndOther_Res	Roof membrane or shingles + roof insulation + wood decking/sheathing + wood framing	U-factor IP	Pre 1978	N/A	Considering 30 years for roof insulation replacement.	0.037	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.037	0.037	0.037	0.037	0.037	0.037
				1978-1997	N/A		0.037	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.037	0.037	0.037	0.037	0.037	0.037
				1998-2007	2001	TABLE 1-I	0.037	0.037	0.051	0.051	0.051	0.051	0.051	0.051	0.051	0.037	0.037	0.037	0.037	0.037	0.037	0.037
				2008-2018	2013	TABLE 140.3-C	0.034	0.028	0.039	0.028	0.039	0.039	0.039	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028
				2019+	2025	TABLE 170.2-A ENVELOPE COMPONENT PACKAGE – Multifamily Standard Building Design	0.028	0.028	0.034	0.028	0.034	0.034	0.039	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028
Windows_ Dwelling Units	FixedAndOperableWindow_Res	CBECS 2018 shows that ~58% of Large Hotels that were built before 2000 are single pane	U factor	Pre 1978	N/A	ASHRAE Handbook of Fundamentals for single pane & T24, 2013 for double pane (58%-42%)	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827
				1978-1997	N/A		0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827	0.827
		Double pane		1998-2007	2001	Table 1-I	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490
				2008-2018	2013	Table 140.3-C	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385	0.385
				2019+	2025	TABLE 170.2-A (High rise residential in multifamily section)	0.325	0.340	0.325	0.325	0.325	0.370	0.370	0.340	0.340	0.340	0.325	0.340	0.325	0.325	0.325	0.325
		CBECS 2018 shows that ~58% of Large Hotels that were built before 2000 are single pane	SHGC	Pre 1978	N/A	ASHRAE Handbook of Fundamentals for single pane & T24, 2013 for double pane (58%-42%)	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559
				1978-1997	N/A		0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559
		Double pane		1998-2007	2001	TABLE 1-I (based on non north with WWR 30-40%)	0.300	0.260	0.260	0.260	0.260	0.310	0.310	0.310	0.310	0.260	0.260	0.260	0.260	0.260	0.260	0.300
				2008-2018	2013	Table 140.3-C	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243	0.243
				2019+	2025	TABLE 170.2-A (High rise residential in multifamily section) Table 110.6-B: Climate zones 1, 3, 5 and 16	0.528	0.228	0.528	0.228	0.528	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.528
Windows_ Nonres (F1)	FixedWindow_NonRes	CBECS 2018 shows that ~58% of Large Hotels that were built before 2000 are single pane	U factor	Pre 1978	N/A	ASHRAE Handbook of Fundamentals for single pane & T24, 2013 for double pane (58%-42%)	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801
				1978-1997	N/A		0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801
		Double pane		1998-2007	2001	Table1-H	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490
				2008-2018	2013	Table 140.3-B	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360
				2019+	2025	Table 140.3-B	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.340	0.360	0.340	0.340	0.340	0.340	0.340	0.360
		CBECS 2018 shows that ~58% of Large Hotels that were built before 2000 are single pane	SHGC	Pre 1978	N/A	ASHRAE Handbook of Fundamentals for single pane & T24, 2013 for double pane (58%-42%)	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563
				1978-1997	N/A		0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563	0.563
		Double pane		1998-2007	2001	Table 1-H (based on non north with WWR 30-40%)	0.430	0.310	0.410	0.410	0.410	0.340	0.340	0.340	0.340	0.310	0.310	0.310	0.310	0.310	0.310	0.430
				2008-2018	2013	Table 140.3-B (100% fixed, 0% operable)	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250
				2019+	2025	Table 140.3-B (100% fixed, 0% operable)	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.220	0.250	0.220	0.220	0.220	0.220	0.220	0.250

APTM41-Envelope (I3:Y37)

It is not clear how the ACH50 value of 3.97 was derived for the 2008-2018 vintage and earlier. It appears that some formulas are present in cells E57:I67, but I was not able to follow the logic to confirm the number. A report focused exactly on the topic of existing multifamily building air leakage may be of interest:

Energy Code Field Studies: Low-Rise Multifamily Air Leakage Testing. 2020

https://www.energycodes.gov/sites/default/files/2021-07/LRMF_Studies_final_report_2020-06-24.pdf

APTM42-Envelope (n/a)

Window shading that is manually operated by the residents is extremely likely in a multifamily building and does have a thermal impact both on heat transfer through windows and on the use of artificial lighting during the day. Most window shading devices are used for two purposes: privacy and light control. Most people lower their shades in the evening so others cannot see into their homes when interior lights are on, and it is dark outside. They are then usually raised during the day unless light is directly bothering an occupant. This type of shade usage is most common in the morning or evening due to low sun angles, but may also occur during midday when watching television. In any of these cases of using shades during the daytime for light control, they are usually applied only to a small number of windows.

APTM43-Envelope (all)

Instead of using a single version of Title 24 to represent a range, perhaps it would make sense to use an average value for all versions of Title 24 within the time period. In other words, for the 1978-1997 range, values from the 1978, 1980, 1982, 1984, 1986, 1987, 1988, 1992, and 1995 versions of Title 24 should all be extracted and averaged together. Currently, for 1978-1997, it uses values from Title 24-1992 only. For the time range 1998-2007, we could average the values from the Title 24 versions from 1998, 2001, and 2005 instead of using just the 2001 values. For the time range 2008-2018, we would average the values from Title 24 versions from 2008, 2010, 2013, and 2016 instead of using just the 2013 values.

APTM44-HVAC System (C6:E6)

How to determine the new construction HVAC system for the single floor that is retail is not clear from Table 1 in the 2025 ACM Reference Manual. There is 17615 ft2 of retail space on the first floor of the building, so one interpretation for a mixed-use building would be that it would represent a single-story building in the table, so it would be "Buildings <= 2 floors," which would be either System 7b SZVAVHP for climate zones 2-15 or System 7c SZVAVDFHP for climate zones 1 or 16. Since the building is five stories and none of the other "retail or grocery store" rows of Table 1 cover that many floors for retail, another interpretation is that it would be the row:

All other spaces in buildings with < 25,000 ft^2 conditioned floor area	Buildings ≤ 3 floors	All	≥ 65 kBtu/h	System 7a – SZVAVAC

Or:

All other spaces in buildings with < 25,000 ft^2 conditioned floor area	4 - 5 floors	All	No limit	System 5 – PVAV

Or:

All other spaces in buildings with 25,000 – 150,000 ft^2 conditioned floor area	≤ 5 floors	All	No limit	System 5 – PVAV

Justifying the interpretation taken in how Table 1 is applied should be described in the scorecard.

APTM45-HVAC System (C5:E5)

For new construction, the residential tenant and common spaces should be separated into two different rows as they are described in Table 26 of the 2025 ACM Reference. Also, the exact type of system depends on the Climate Zone. For dwelling units in buildings with more than 3 habitable floors, Climate Zones 2-15, it is “Single zone system with constant volume fan, no economizer, DX heat pump with electric resistance supplemental heat,” but for Climate Zones 1 and 16 it is “Single zone system with constant volume fan, no economizer, DX heat pump with gas furnace supplemental heat.” They are the same, but the type of supplemental heat differs. Similar differences also apply for common areas, except they do not say “no economizer.”

APTM46-HVAC System (C17)

No heating efficiency metrics are shown for new construction for non-residential spaces.

APTM47-HVAC System (C12:D15)

The existing vintages should be separated into the residential, common, and non-residential retail equipment to correspond to the categories used in the first table on the tab. For example, boiler efficiencies are missing.

APTM48-Ventilation (F9:F12)

The cfm/ft2 for retail is 0.2, not 0.15, based on Table 120.1-A.

TABLE 120.1-A– Continued Minimum Ventilation Rates

Occupancy Category - *Retail*	Minimum Occupant Load Density (persons / 1000 ft²)	Area-based Minimum Ventilation R_a (cfm/ft²)	Air Class	Notes
Sales (except as below)	17	0.2	2	--
Mall common areas	17	0.15	1	F
Barbershop	17	0.4	2	--
Beauty and nail salons	17	0.4	2	--
Pet shops (animal areas)	17	0.2	2	--
Supermarket	17	0.2	1	F
Coin-operated laundries	17	0.3	2	--

APTM49-Ventilation (D39, F39)

A parking garage only has an exhaust rate of 0.75, so the 15 cfm/person and 0.15 cfm/ft2 should be removed from that row.

Parking garages	-	0.75	2	C

APTM50-Ventilation (Column E)

For residential ventilation, the formula referenced is.

Total Required Ventilation Rate [ASHRAE 62.2:4.1.1].

$Q_{tot} = 0.03A_{floor} + 7.5(N_{br} + 1)$ (Equation 160.2-B)

WHERE:

Q_{tot} = total required ventilation rate, cfm

A_{floor} = dwelling-unit floor area, ft^2

N_{br} = number of bedrooms (not to be less than 1)

But the formula in the cell points to the Zone tab, column E, which is the number of dwelling units in the zone and is not the number of bedrooms. The number of bedrooms is shown in column F (it is text, so it would need to be converted into a number). In addition, the total of the formula should be multiplied by column E since that is the number of dwelling units represented by the zone, as well as a floor multiplier when appropriate.

APTM51-Ventilation (Column G for residential spaces)

According to 160.2(b)(2)(vi), local mechanical exhaust is also required for kitchens and bathrooms.

APM52-Ventilation (n/a)

Operable windows are present in the building for dwelling units and consist of 25% of the windows. Based on many studies, many people will use their operable windows to provide natural ventilation. This will reduce the amount of HVAC energy used, especially cooling, during outdoor conditions that people perceive as favorable. See:

- Assessing the impact of occupant behavior on residential building performance: A case study of window operation by Farjadnia et al.
- 2014 Building America House Simulation Protocols by Wilson et al. Section 4.2.1
- DEER Residential EnergyPlus Prototypes: Development & Calibration, Section 2.2.8
- Low-Rise Multifamily Prototype Model Validation by Mohammad Dabbagh et al.

People value fresh air, and in my experience tend to leave windows open, at least to a small amount, in even a broader range of outdoor conditions than stated, 55°F - 75°F. Wind speed and humidity are two factors that should be considered. Adapted comfort, as described in ASHRAE Standard 55, should be considered. I believe many people tend to find high humidity conditions, even at 75°F, a reason not to use natural ventilation. A cool summer morning is likely to spur the use of opening windows for natural ventilation, which will probably not be closed until the house is slightly more uncomfortable than usual by the afternoon; then windows will be shut, and air-conditioning will be used. Then, cool evenings will often prompt the reopening of windows as people get home. This behavior is again very dependent on whether the apartment is likely to be occupied during the day, when a person may close the windows. An additional issue is that people close windows for security reasons when they know they will be away from home. This points to having separate representations of daytime occupied and unoccupied apartments in the prototypes.

APTM53-Thermostat (A3:A6)

The setpoints are described as “school.”

APTM54-Thermostat (A1:R1, B4:Q4)

The top table is described as “Thermostat Setpoints - ApartmentMidRise: ResidentialLiving & ResidentialCommon,” but this should be separated into two tables since ResidentialLiving and ResidentialCommon should have different heating setback values, 60F and 68F, respectively.

APTM55-Thermostat (n/a)

A few surveys found online, “surveys of thermostat settings,” should be reviewed:

https://joybird.com/blog/the-great-thermostat-debate/

https://www.angi.com/articles/average-room-temperature.htm

https://www.oklahoman.com/story/news/2025/07/03/what-temperature-should-i-set-my-thermostat-according-to-oklahomans/84444431007/

https://www.appriseinc.org/wp-content/uploads/2022/09/Residential-Thermostat-Settings-and-Temperature-Preferences-Study-9-10-21.pdf

APTM56-Thermostat (n/a)

For apartments, the thermostat setting is a complicated topic. Using a 68°F constant setpoint for heating and 78°F for cooling is a simple approach, but I would recommend a more comprehensive approach for the prototype. People have a diversity of temperatures that they find comfortable. To try to capture this effect, use several different schedules for apartments that have a wide diversity of values, including deep setbacks for unoccupied rooms.

APTM57-Water Heater (A5:D5)

The choice of two options is made in reference to this section “Title 24 2025: §170.2(d)3: Gas Storage with recirculation and solar water heating system”:

B. For gas or propane systems serving multiple dwelling units, the water-heating system that includes the following components shall be installed:

   i. For Climate Zones 1 through 9, gas service water-heating systems with a total installed gas water-heating input capacity of 1 MMBtu/h or greater shall have gas service water-heating equipment with a minimum thermal efficiency of 90 percent. Multiple units are allowed to meet this requirement with an input capacity-weighted average of at least 90 percent.

      Exception 1 to Section 170.2(d)Bi: Individual gas water heaters with input capacity at or below 100,000 Btu/h shall not be included in the calculations of the total system input or total system efficiency.
      Exception 2 to Section 170.2(d)Bi: If 25 percent of the annual water-heating requirement is provided by site-solar energy or site-recovered energy.

   ii. A solar water-heating system meeting the installation criteria specified in Reference Residential Appendix RA4 and with a minimum solar savings fraction of either a. or b. below:

      a. A minimum solar savings fraction of 0.20 in Climate Zones 1 through 9 or a minimum solar savings fraction of 0.35 in Climate Zones 10 through 16; or

      b. A minimum solar savings fraction of 0.15 in Climate Zones 1 through 9 or a minimum solar savings fraction of 0.30 in Climate Zones 10 through 16. In addition, a drain water heat recovery system that is field verified as specified in the Reference Appendix RA3.6.9.

It is not clear whether a 90% boiler would not be more cost-effective in Climate Zones 1 through 9.

APTM58-Water Heater (E11, E14, E30, E47)

Several zones that are not dwelling units have the wrong values shown for hot water consumption.

Zone	Value Shown	Expected Value
F1 CORRIDOR	0.18	0.00
F1 LOUNGE	0.18	0.09
F2-4 LAUNDRY	0.00	0.578
F5 LAUNDRY	0.18	0.578

APTM59-Water Heater (E20:E28, E31:E41, E44:E45)

The residential dwelling unit hot water consumption per person is not consistent with values shown as 0.00, 0.09, 0.18 depending on the row. The cited source for this information is T24, 2022, Appendix 5.4A, sheet: “2022 SpaceFuncData-Input,” but that cell shows “RULE Spc:DwellingUnitXGalPerDay” instead of a specific value. It is probably a link error. Whatever value, it should be consistent based on the dwelling unit size or number of people.

APTM60-Water Heater (F9:F46)

Given the problems with the number of people (see APTM15-Zones, APTM16-Zones), the calculated values are probably incorrect.

APTM61-Water Heater (B51:B52, F51:F52)

The total peak hot water usage of 116 gal/hr seems too small for a five-story mixed residential and retail building of 141,014 ft2, which probably has about 170 people, as does the computed required storage volume of 46 or 47 gallons, considering that a single-family home usually needs a 40-gallon water heater.

APTM62-Water Heater (B54, B64)

The table of numeric assumptions for calculating the values in the "Water Heater Capacity, Tank Capacity, and Tank Loss" is repeated twice below that table. The values are generally the same except the Design supply temperature, which shows as 135F or 130F depending on the table.

APTM63-Water Heater (n/a)

Given the different rules in the ACM related to multifamily domestic hot water, it is surprising that none of the assumptions used are based on Section 6.11 of the ACM.

APTM64-Water Heater (n/a)

Following the approach used in "Multifamily Building Data Analysis in Support of ASHRAE Standard 140 Test Case Development" by Karpman Consulting 2025, each occupant consumes 3.73 gallons/hour. Assuming 176 people for the residential units, that is 656 gallons/hour, or 10.9 gallons/minute. This approach is based on EPA Energy STAR simulation guidelines. Another source of data on this is "2014 Building America House Simulation Protocols" by Wilson et.al., Section 2.14.

APTM65-Water Heater (B55, B65)

The cold-water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as for other prototypes.

APTM66-Interior Lights (B7)

For office areas greater than 250 ft2, the value should be 0.60, not 0.65 W/ft2.

Office Area > 250 square feet	0.60	Decorative/Display and Portable lighting for office areas[5]	0.20
Office Area ≤ 250 square feet	0.65	Decorative/Display and Portable lighting for office areas[5]	0.20

APTM67-Interior Lights (B12)

For parking garages, some of the area should be “daylight adaptation zones,” which can be a significant portion of the total area since it can be within 66 ft of the entrance (see note 3 of the referenced table).

Parking Garage Area Parking Zone and Ramps	0.10	First ATM or Ticket Machine	100 W
Parking Garage Area Parking Zone and Ramps	0.10	Additional ATM or Ticket machine	50 W each
Parking Garage Area Daylight Adaptation Zones[3]	1.00	-	-

APTM68-Interior Lights (Columns C and F)

The column labeled “Additional LPD for Portable Lighting (W/ft2)” is shown as NA but should cover any Additional Lighting Power Allowance and should have a value where appropriate. Meeting rooms, lobbies, corridors, and offices are likely to take advantage of these credits in order to try to make the general building look more appealing to prospective tenants:

Conference, Multipurpose and Meeting Area	0.75	Display	0.25
Conference, Multipurpose and Meeting Area	0.75	Wall Display MH <= 10'6"	2 W/ft
Conference, Multipurpose and Meeting Area	0.75	Wall Display MH 10'7" to 14'	2.35 W/ft
Conference, Multipurpose and Meeting Area	0.75	Wall Display MH > 14'	2.66 W/ft
Conference, Multipurpose and Meeting Area	0.75	Floor Display & Task MH <= 10'6"	0.30
Conference, Multipurpose and Meeting Area	0.75	Floor Display & Task MH 10'7" to 14'	0.35
Conference, Multipurpose and Meeting Area	0.75	Floor Display & Task MH > 14'	0.40
Corridor Area	0.40	Decorative/Display	0.25
Lobby, Main Entry	0.70	Display	0.25
Lobby, Main Entry	0.70	Wall Display MH <= 10'6"	3 W/ft
Lobby, Main Entry	0.70	Wall Display MH 10'7" to 14'	3.5 W/ft
Lobby, Main Entry	0.70	Wall Display MH > 14'	4 W/ft
Office Area > 250 square feet	0.60	Decorative/Display and Portable lighting for office areas[5]	0.20
Office Area ≤ 250 square feet	0.65	Decorative/Display and Portable lighting for office areas[5]	0.20

In addition, the retail areas are even more likely to take advantage of this (from Table 140.6C):

Retail Sales Area	Retail Merchandise Sales	0.95	Decorative	0.35
Retail Sales Area	Retail Merchandise Sales	0.95	Wall Display MH <= 10'6"	9.5 W/ft
Retail Sales Area	Retail Merchandise Sales	0.95	Wall Display MH 10'7"-14'	11.2 W/ft
Retail Sales Area	Retail Merchandise Sales	0.95	Wall Display MH > 14'	12.7 W/ft
Retail Sales Area	Retail Merchandise Sales	0.95	Floor Display & Task MH <= 10'6"	0.45
Retail Sales Area	Retail Merchandise Sales	0.95	Floor Display & Task MH 10'7"- 14'	0.52
Retail Sales Area	Retail Merchandise Sales	0.95	Floor Display & Task MH > 14'	0.60
Retail Sales Area	Retail Merchandise Sales	0.95	Valuable Display Case	0.50
Retail Sales Area	Retail Merchandise Sales	0.95	General Lighting in the enclosed space of ceiling height > 10'	0.10

This comment also applies to the existing vintage from Title 24-2016, which also has additional lighting power allowances.

APTM69-Interior Lights (D10)

The source for the retail value should be “T24, 2025, Table 140.6-C”.

APTM70-Interior Lights (B13, D13, A15)

The “High-Rise Residential Living Spaces” LPD is one of the most important values from this table, and the derivation cites the single-family ACM Appendix E, Tables 2 and 8, and the equation from Table 2 is shown as Annual kWh=100+0.1775×CFA (ft^2)(Eq. 1), but the translation of that to an LPD of 0.79 is non-obvious and should be explained in detail. Another possible source for this type of information could be Section 2.17 of “2014 Building America House Simulation Protocols” by E. Wilson et al.

APTM71-Interior Lights (G4:G12)

Assuming that the lighting for existing buildings is compliant with Title 24-2016 seems very optimistic for many multifamily buildings, especially those traditionally marketed to low-income tenants or in areas that are disadvantaged.

APTM72-Interior Lights (E4, E7, E9:E11)

Several existing space types have the wrong values shown for the LPD:

Zone	Value Shown	Expected Value
Convention, Conference, Multipurpose and Meeting Area	0.85	1.2
Office Area (>250 square feet)	0.65	0.75
Electrical, Mechanical, Telephone Rooms	0.4	0.55
Retail Sales Area (Retail Merchandise Sales)	1	1.2
Laundry Area	0.45	0.7

APTM73-Ext Lights (C11)

The amount of parking area calculated does not deduct the amount of parking in the garage on the lower level. The value of 84,136 ft2 calculated in this cell should be reduced by the

area of the garage.

APTM74-Ext Lights (B11:C11)

It is not clear where the parking area assumption came from

> "1.5 parking space per living unit for residential
>
> 1 parking space per 360 ft^2 of gross building area"

This does not seem to be described in either of the sources cited.

In addition, the formula for the C11 cell:

> =400*B25*1.5+B23/B24/360*400

is

> =400*NumOfDwellingUnits*1.5+TotalFloorArea/NumFloorsAboveGrade/360*400

Includes two things that look suspicious

- The two portions of the formula are added together, which seems to be quite a few parking spaces.
- The second portion of the formula divides the floor area by the number of floors, which would represent not the gross building area but the area of a single floor.

APTM75-Ext Lights (E15)

The lighting power allowance factor for the sales canopy is shown as 0.137 W/ft2 for lighting zone 2. This is appropriate for a non-sales canopy, so either the number is wrong or the description of the row is wrong.

Lighting Application	Lighting Zone 0	Lighting Zone 1	Lighting Zone 2	Lighting Zone 3	Lighting Zone 4
Sales Canopies. Allowance for the total area within the drip line of the canopy. Luminaires qualifying for this allowance shall be located under the canopy.	Not applicable	No Allowance	0.470 W/ft²	0.622 W/ft²	0.740 W/ft²
Non-sales Canopies and Tunnels. Allowance for the total area within the drip line of the canopy or inside the tunnel. Luminaires qualifying for this allowance shall be located under the canopy or tunnel.	Not applicable	0.057 W/ft²	0.137 W/ft²	0.270 W/ft²	0.370 W/ft²

APTM76-Ext Lights (B15:C15)

The sales canopy area formula in cell C15 divides the floor area by the number of above-ground stories and then multiplies by the 0.1 factor. It is not clear why it is being divided by the number of stories, which results in using a base area of a single floor rather than the total building gross area.

APTM77-Ext Lights (B12:C12)

Having only three entrances seems a bit on the low side for the size of the building 141,000 ft2 and being five stories, especially with the first floor being primarily retail. The statement in cell B12:

> "3 building entrances for NR (1 per 10,000 ft^2 footprint)"

Is not consistent with the floor space nor with this being mostly a multifamily building (assuming NR stands for non-residential).

In addition, each of the retail spaces is likely to have an entrance. There are 12 retail spaces. Looking at the first-floor floor plan on the Zones tab implies that there are also probably five entrances for the families in the building: facing north between the stores, near each flight of stairs, near the lounge/rec room on the east side of the building, and near the leasing office on the west side of the building.

APTM78-Ext Lights (B14)

The building façade factor of 0.5 described in cell B14:

> 0.1 W per ft^2 of facade area
>
> Façade Area = Perimeter × 0.5 × 30 ft (30 ft is the max height that is lit) *
>
> The 0.5 factor assumes that only half of the perimeter walls are lit.
>
> -Formula is developed based on the 2019 Outdoor Lighting CASE report

The referenced table shows:

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications											
		Applied to % of Building S.F. in Category									
Lighting Allowance	Assumptions	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other
Building Facades	**30' building height, 2 floors per building** (20% of applicable facades are lit)	25%	50%	50%	25%		50%	25%	25%	5%

But this table does not have a value for a mixed multifamily and retail building, so I believe the value being taken is the 50% for retail, but that implies that the second floor of the building is also retail, which it is not. If multifamily is "other," then perhaps 27.5% would make more sense for the multiplier.

In general, the multipliers that are from Table 36 do not seem to be getting applied in other rows of the scorecard.

APTM79-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as provided for other prototypes.

APTM80-Equipment (D3:D11)

The source cited is "Plug load and schedules project with CEC, 2025," which is probably a shortening of the title "Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Non-residential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual," which was published in August 2025. The equipment power densities from that report are shown as:

Table 1: Proposed New EPD Values, by Space Type

Space Type	Current EPD[1] (W/ft²)	Proposed EPD (W/ft²)
Office Area (>250 square feet)	1.5	0.7
Office Area (<250 square feet)	1.5	0.7
Kitchen/Food Preparation Area	1.5	1.5[2]
Kitchenette or Residential Kitchen	1.0	1.0[3]
Retail Sales Area (Grocery Sales)	1.0	0.4[4]
Retail Sales Area (Retail Merchandise Sales)	1.0	0.4
Retail Sales Area (Fitting Room)	1.0	0.4
High-Rise Residential Living Spaces	-[5]	Reduced by 3.5%
Hotel/Motel Guest Room	0.5	0.5[6]
Classroom, Lecture, Training, Vocational Areas[7]	1.0	0.8

But none of these values seem to be getting used in the scorecard. Only two of the values, "office area" and "retail merchandise sales," correspond with values shown in the scorecard based on this source.

APTM81-Equipment (C9:C10)

Presuming that the source of the plug load power density is the Title 24-2025 spreadsheet called "Appendix_5.4A_SpaceUseData-T24N.xlsx," then almost all the values are correct except Retail Sales Area (Retail Merchandise Sales) and Laundry Area.

APTM82-Equipment (B13:C13)

Residential plug loads are difficult to estimate, and if another source is sought to inform the values used, see Sections 2.1.8 and 3.1.8 of "2014 Building America House Simulation Protocols" by Wilson et al.

APTM83-Equipment (D13)

There doesn't appear to be a source called "Plug load and schedules project with CBECC-COM, 2019".

APTM84-Equipment (D18, A22:F25)

The number of elevators, five, is based on

https://buildingandinteriors.com/find-the-magic-number-elevators-per-building/

That URL no longer works, but it can be found here:

https://web.archive.org/web/20250704165142/https://buildingandinteriors.com/find-the-magic-number-elevators-per-building/

In addition, the article has been updated from December 2023 to September 2025 here:

https://buildingandinteriors.com/number-of-elevators-required-in-a-building/

Both provide the same information:

No. of Elevators required in a residential building (Apartment / Condo / Dormitory):

1. There should be one elevator for every 90 units. A maximum of 150 feet from the elevators to the farthest elevator is a must.
2. City apartments and really high priced units can require an elevator for every 50 – 60 units.
3. Always have an elevator with higher ceilings (9 feet) and higher weight capacity (3500 lbs) for moving furniture and heavy items. If you are over 10 floors, a service elevator or small freight elevator should be considered.
4. Four stories or more above grade a gurney sized elevator is required.

This is not the rule of thumb used in the scorecard, which is:

> One elevator for approximately 50,000 ft^2 in use.
>
> For every two floors or two and a half floors, one more elevator was added.

These are the rules of thumb from this same article for office buildings.

The drawing on the Zones tab shows three elevators.

APTM85-Equipment (C18, A26:F28)

The design load for elevators 4361 W is not consistent with the 50 kW assumption used under the table to justify it. It is not clear where the 10 kW from ACM 2022 comes from.

APTM86-Schedules (Column D)

Why is “T24 2022 - Appendix 5.4B” the source when an updated spreadsheet is available with Title 24-2025?

APTM87-Schedules (F22:K24, AB22:AC24)

The values shown for the heating setpoint for residential living are 68F for all hours, but the Appendix 5.4B schedule that corresponds with this has a setback to 60F for hours 1 to 6 and 23 to 24.

APTM88-Schedules (Y73, AA73)

Two values in the retail electric equipment (plug load) schedule are not consistent with the source. For Saturday hours, 20 and 22 have values of 0.60 and 0.40, respectively, in the scorecard, but in the source "Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Non-residential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual," those values are 0.90 and 0.60, respectively.

APTM89-Schedules (n/a)

Given the variation of schedules of individuals and families, the use of stochastic schedules may be appropriate for each individual apartment in a multifamily building. See Section 3.2.1, Appendix D, and Appendix E of "End-Use Load Profiles for the U.S. Building Stock Methodology and Results of Model Calibration, Validation, and Uncertainty Quantification" (March 2022) NLR 80889. In addition, the draft CalBEM residential prototype documentation showed the intention to use DHW schedules that are stochastic.

APTM90-Schedules (n/a)

The occupancy schedule for the apartments seems overly simple. Apartments that are empty during the day (all occupants having left) have very different operations than when there are always occupants (a stay-at-home parent or work-at-home occupant). At least some random schedule for occupancy should be used to capture this variation. The occupancy, in turn, should determine when appliances are used. A stochastic approach might make sense for occupancy, since it is a major driving force for loads in buildings, with the heat from occupancy and appliances operated by occupants. The schedule is the same for weekdays and weekends, which seems very unlikely. Perhaps a graphical description of the intended schedules would be appropriate. Combining stochastic scheduling for each occupant's arrival and departure times might make sense. I believe there have been studies of the variation in activities done by people that might be able to help drive this process.